

General Chemistry (CHEM 1310)

Midterm Examination 3

Name	GT ID (90...)
Section (A or B)	

Indicate your response to each question in the box or on the line(s) provided. For multiple-choice questions, fill in the circle next to your choice clearly and completely. For multiple-answer questions, fill in the circles completely next to statements you wish to select.

Academic Integrity Guidelines

1. This quiz is closed note, closed book, and closed Internet. You are permitted only a single crib sheet and a calculator lacking Internet connectivity.
2. You may not access any other resource or website aside from this document for the duration of the exam.
3. You may not collaborate with any other person (in person or electronically) on any portion of the exam, including general discussion of exam question topics prior to exam submission by all parties.
4. You are not permitted to screen-capture or otherwise record the questions on this exam.
5. All work submitted for this exam must be your own.

I affirm that I have abided by all academic integrity guidelines as listed above.

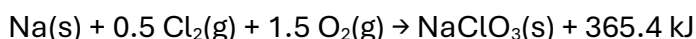
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Do not open this exam until you are instructed to do so.

1. **5 points.** One mole of an ideal gas with heat capacity $0.205 \text{ L}\cdot\text{atm}/^\circ\text{C}$ is heated at a constant pressure of 0.980 atm from 20.0°C to 95.0°C . The gas expands from a volume of 0.500 L to 4.800 L . What is the change in internal energy (ΔU) of the gas for this process?

- A. ☐ $+15.4 \text{ L}\cdot\text{atm}$
 B. ☐ $+11.2 \text{ L}\cdot\text{atm}$
 C. ☐ $-11.2 \text{ L}\cdot\text{atm}$
 D. ☐ $+19.6 \text{ L}\cdot\text{atm}$
 E. ☐ $-1.13 \text{ L}\cdot\text{atm}$

2. **5 points.** Formation of solid sodium chlorate from its elements is an exothermic process with ΔH° equal to -365.4 kJ/mol .



In one preparation of NaClO_3 using this reaction, 80.2 kJ of heat were evolved. How many moles of NaClO_3 were produced?

mol

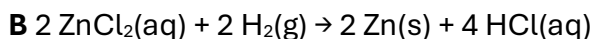
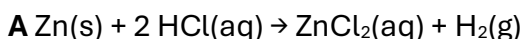
3. **5 points.** Use the standard enthalpies of formation given to determine the standard enthalpy of reaction for the process below.



Substance	$\Delta H^\circ_f \text{ (kJ/mol)}$
$\text{CaCO}_3(\text{s})$	-1207
$\text{HCl}(\text{aq})$	-167
$\text{CaCl}_2(\text{aq})$	-877
$\text{CO}_2(\text{g})$	-394
$\text{H}_2\text{O}(\text{l})$	-286

- A. ☐ -350 kJ
 B. ☐ -270 kJ
 C. ☐ -1557 kJ
 D. ☐ $+1541 \text{ kJ}$
 E. ☐ -16 kJ

4. **5 points.** Consider reactions **A** and **B** below. How is the enthalpy of reaction **B** related mathematically to the enthalpy of reaction **A**?



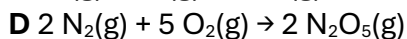
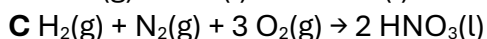
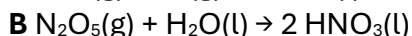
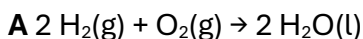
- A. ☐ $\Delta H^\circ(\text{B}) = -2\Delta H^\circ(\text{A})$
 B. ☐ $\Delta H^\circ(\text{B}) = -\Delta H^\circ(\text{A})$
 C. ☐ $\Delta H^\circ(\text{B}) = +2\Delta H^\circ(\text{A})$
 D. ☐ $\Delta H^\circ(\text{B}) = +3\Delta H^\circ(\text{A})$
 E. ☐ The enthalpies are not related at all.
5. **6 points.** Three metals **A – C** with equal mass are placed on a hot plate, which provides a constant amount of heat to each metal. Initial and final temperatures of each metal are below. Rank the metals from smallest (1) to largest (3) specific heat.

Metal	Initial Temperature (°C)	Final Temperature (°C)
A	20.0	38.2
B	20.0	47.1
C	20.0	62.3

smallest c

largest c

6. **5 points.** Which of the following equations shows how reactions **A – C** can be combined to generate reaction **D**?



- A. ☐ $\text{D} = \text{A} + 2\text{B} - 2\text{C}$
 B. ☐ $\text{D} = \text{A} + \text{B} - 2\text{C}$
 C. ☐ $\text{D} = -2\text{A} - \text{B} + 2\text{C}$
 D. ☐ $\text{D} = -\text{A} - 2\text{B} + 2\text{C}$
 E. ☐ $\text{D} = \text{A} - \text{B} - \text{C}$

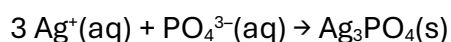
7. **6 points.** Indicate the expected sign of the standard entropy ΔS° for each process below.

[☐ Positive ☐ Negative] $\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2 \text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$

[☐ Positive ☐ Negative] $\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightarrow 2 \text{NH}_3(\text{g})$

[☐ Positive ☐ Negative] $\text{CO}_2(\text{s}) \rightarrow \text{CO}_2(\text{g})$

8. **6 points.** The process below occurs spontaneously from standard conditions. Indicate whether each statement below is true or false.



[☐ True ☐ False] For this process, $\Delta H^\circ_{\text{sys}}$ is less than zero.

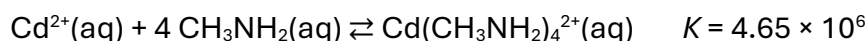
[☐ True ☐ False] The free energy of the product is less than the sum of the free energies of the reactants.

[☐ True ☐ False] The fact that the reaction is spontaneous implies that $\Delta S^\circ_{\text{sys}}$, the entropy change of the reaction system, is greater than zero.

9. **5 points.** For which of the following combinations of signs of ΔH° and ΔS° is a reaction spontaneous in the forward direction from standard conditions only *above* a cutoff temperature?

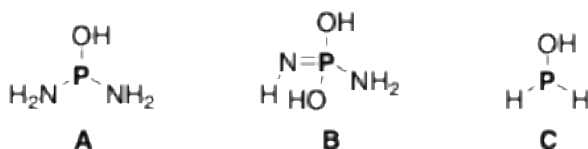
- A. ☐ The temperature dependence of spontaneity cannot be determined from ΔH° and ΔS° alone.
- B. ☐ $\Delta H^\circ > 0$, $\Delta S^\circ < 0$
- C. ☐ $\Delta H^\circ > 0$, $\Delta S^\circ > 0$
- D. ☐ $\Delta H^\circ < 0$, $\Delta S^\circ < 0$
- E. ☐ $\Delta H^\circ < 0$, $\Delta S^\circ > 0$

10. **5 points.** Consider the process below. What is ΔG° for this process at 293 Kelvin? Enter your response with appropriate sign in kilojoules per mole (kJ/mol) to the nearest 0.1 kJ/mol. The ideal gas constant R is 8.314 J/K·mol.



kJ/mol

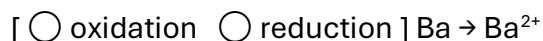
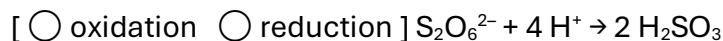
11. **6 points.** Rank the compounds below from most negative to most positive oxidation number at the bolded (phosphorus) atom. "R" is a generic carbon group (that is, carbon is linked to the indicated atom).



most negative ON

most positive ON

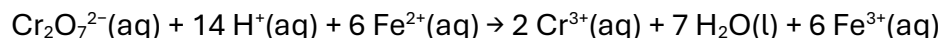
12. **6 points.** Consider the two half-reactions shown below. Indicate whether each is an oxidation or reduction process as written (electrons are omitted).



When combining the half-reactions above to create a balanced redox reaction with the smallest whole-number coefficients, how many electrons are transferred in the reaction?

[☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6]

13. **5 points.** Identify the oxidizing agent and reducing agent in the balanced redox reaction below.



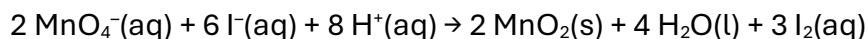
$\text{Cr}_2\text{O}_7^{2-}$ is the [☐ oxidizing agent ☐ reducing agent].

Fe^{2+} is the [☐ oxidizing agent ☐ reducing agent].

In this reaction, the protons (H^+) undergo

[☐ oxidation ☐ reduction ☐ no change in oxidation state].

14. **5 points.** For the reaction below, E° is +1.15 V. Fill in the blanks below.



As written and starting from standard conditions, the reaction is...

[☐ spontaneous ☐ non-spontaneous].

ΔG° for the reaction is... [☐ positive ☐ negative ☐ zero].

K for the reaction is...

[☐ greater than 1 ☐ less than 1 ☐ equal to 1].

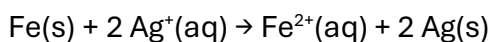
15. **5 points.** Which of the following metals will dissolve spontaneously in 1.0 M nitric acid (HNO_3)? In other words, for which of the metals is the reaction below spontaneous from standard conditions? Use the standard reduction potentials provided. Select all that apply.



Half-reaction	E°_{red} (V)
$\text{Li}^+ + \text{e}^- \rightarrow \text{Li}(\text{s})$	-3.04
$\text{Ba}^{2+} + 2 \text{e}^- \rightarrow \text{Ba}(\text{s})$	-2.91
$\text{Pd}^{2+} + 2 \text{e}^- \rightarrow \text{Pd}(\text{s})$	+0.915
$\text{Cs}^+ + \text{e}^- \rightarrow \text{Cs}(\text{s})$	-3.026
$\text{Re}^{3+} + 3 \text{e}^- \rightarrow \text{Re}(\text{s})$	+0.500

- A. ☐ Li
- B. ☐ Ba
- C. ☐ Pd
- D. ☐ Cs
- E. ☐ Re

16. **5 points.** A voltaic cell is constructed using iron metal, 1.00 M $\text{Fe}(\text{NO}_3)_2$, silver metal, and 1.00 M AgNO_3 . The salt bridge contains $\text{KNO}_3(\text{aq})$. The spontaneous redox reaction that occurs in the cell is below. Fill in the blanks to describe this cell.



$\text{Fe}(\text{s})$ and $\text{Fe}^{2+}(\text{aq})$ are found in the [☐ cathode ☐ anode].

$\text{Ag}(\text{s})$ and $\text{Ag}^+(\text{aq})$ are found in the [☐ cathode ☐ anode].

Electrons flow from [☐ Fe ☐ Fe^{2+} ☐ Ag ☐ Ag^+]

to [☐ Fe ☐ Fe^{2+} ☐ Ag ☐ Ag^+] as the cell discharges.

As the cell discharges, K^+ ions migrate toward the [☐ cathode ☐ anode] and NO_3^- ions migrate toward the [☐ cathode ☐ anode].

17. **5 points.** Which of the following equations involving a thermodynamic state or path function is *NOT* correct?

A. ☐ $H = U + PV$

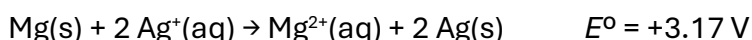
B. ☐ $w = -P \Delta V$

C. ☐ $\Delta G = -E/nF$

D. ☐ $q = C \Delta T$

E. ☐ $G = H - TS$

18. **5 points.** Consider a galvanic cell based on the following spontaneous redox reaction under standard conditions. Indicate whether each intervention (on the standard cell, changing nothing else) will cause an increase, decrease, or no change in the cell potential.



[☐ increase ☐ decrease ☐ no change] Increasing the molarity of Ag^+

[☐ increase ☐ decrease ☐ no change] Increasing the molarity of Mg^{2+}

[☐ increase ☐ decrease ☐ no change] Increasing the temperature

19. 5 points. A current of 0.100 A was applied for 1.00 hour for the electrolytic oxidation of an unknown metal to aqueous metal cation. During this process, 1.24×10^{-3} mol of the metal cation were produced. What is the charge on the metal cation? Faraday's constant is 96485 C/mol.

- A. ☐ +1
- B. ☐ +2
- C. ☐ +3
- D. ☐ +4
- E. ☐ +5